

Network Slicing in 5G New Radio: Architecture, Mechanisms, and Use Cases

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Abstract—Network slicing is a pivotal enabling technology in 5G New Radio (NR) that permits the logical partitioning of a shared physical infrastructure into multiple isolated, end-to-end virtual networks, each tailored to specific service requirements. This paper presents a comprehensive study of network slicing in 5G NR, covering its architecture, key enabling technologies—including Software-Defined Networking (SDN), Network Functions Virtualization (NFV), and Multi-Access Edge Computing (MEC)—slice lifecycle management, and representative use cases such as enhanced Mobile Broadband (eMBB), Ultra-Reliable Low-Latency Communications (URLLC), and massive Machine-Type Communications (mMTC). Performance trade-offs and open challenges are also discussed.

Index Terms—network slicing, 5G NR, SDN, NFV, MEC, eMBB, URLLC, mMTC, quality of service, resource allocation.

I. INTRODUCTION

The fifth generation of mobile networks (5G New Radio) represents a fundamental paradigm shift from its predecessors, not merely in terms of raw throughput but in its unprecedented ability to simultaneously support a highly heterogeneous mix of services with vastly different and often conflicting requirements [1]. Traditional mobile network architectures are built around a single, monolithic design that delivers a uniform connectivity model to all users and applications.

This one-size-fits-all approach is fundamentally ill-suited to the diversity of 5G service demands, which span a range from multi-gigabit-per-second video streaming, to sub-millisecond industrial control loops, to low-power sensor networks connecting billions of devices. Serving all these requirements on a single network design inevitably forces operators to over-provision resources and sacrifice efficiency.

Network slicing addresses this challenge by enabling mobile network operators to instantiate multiple logically isolated, end-to-end virtual networks—known as *network slices*—atop a common, shared physical infrastructure [2]. Each slice is allocated a dedicated, guaranteed set of virtualised resources and is governed by service-specific policies that ensure the required Quality of Service (QoS) for its target application class. The 3GPP specifications for 5G NR, beginning with Release 15, formally incorporate network slicing as a core first-class architectural feature of the system.

This paper provides a structured and comprehensive overview of network slicing in 5G NR. Section II reviews the core enabling technologies. Section III details the 5G NR slicing architecture as defined by 3GPP. Section IV discusses the slice lifecycle management framework. Section V presents three representative use cases alongside a comparative QoS performance analysis. Section VI identifies open research challenges, and Section VII concludes the paper.

II. ENABLING TECHNOLOGIES

Network slicing in 5G NR is made possible by the convergence of three foundational technologies: Software-Defined Networking (SDN), Network Functions Virtualization (NFV), and Multi-Access Edge Computing (MEC). Each contributes a distinct capability to the overall slicing framework.

A. Software-Defined Networking

SDN decouples the network control plane from the data plane, centralising forwarding intelligence in a logically centralised software

controller [3]. In the context of network slicing, SDN enables fine-grained and dynamically programmable control over traffic flows across heterogeneous network domains—including the Radio Access Network (RAN), the transport network, and the core. This programmability is the foundation upon which per-slice traffic isolation and customised routing policies are enforced. The OpenFlow protocol and its successors have been instrumental in realising SDN-based slicing in both research testbeds and commercial deployments.

B. Network Functions Virtualization

NFV abstracts network functions from their traditional dependence on proprietary, dedicated hardware appliances, re-implementing them as software-based Virtual Network Functions (VNFs) executing on commodity server infrastructure [4]. For network slicing, NFV is essential in composing the slice-specific service function chains—ordered sequences of VNFs such as firewalls, traffic shapers, and policy enforcement points—from a shared, elastic pool. This enables rapid slice provisioning, flexible scaling in response to dynamic traffic demand, and efficient resource utilisation through statistical multiplexing of compute resources across slices.

C. Multi-Access Edge Computing

MEC relocates compute, storage, and networking resources from centralised cloud data centres to the network edge, in close physical proximity to end devices and base stations [5]. When integrated with network slicing, MEC provides latency-sensitive slices—particularly URLLC slices—with the ability to process traffic locally at the edge, meeting stringent end-to-end delay budgets that are unattainable through communication with a geographically distant central cloud. MEC also enables context-aware service personalisation by exposing radio network information to edge-hosted application servers.

III. 5G NR NETWORK SLICING ARCHITECTURE

The 3GPP 5G system architecture, formally specified in 3GPP TS 23.501, defines network slicing across three principal domains: the 5G Core Network (5GC), the Next-Generation Radio Access Network (NG-RAN), and the transport network interconnecting them. Together, these domains form an end-to-end slice that spans from the air interface to the operator's data network and beyond.

Each network slice is uniquely identified by a Single Network Slice Selection Assistance Information (S-NSSAI) element, which is composed of a Slice/Service Type (SST) indicating the intended service category, and an optional Slice Differentiator (SD) that allows operators to distinguish between multiple slice instances of the same

type. The Access and Mobility Management Function (AMF) and the Network Slice Selection Function (NSSF) collaborate during UE registration and session establishment to associate each UE session with the correct slice or set of slices.

Figure 1 illustrates the layered network slicing architecture. The physical infrastructure layer provides the shared pool of compute, storage, and networking hardware. The NFV/SDN virtualisation layer abstracts this hardware into virtualised resources. The three standardised slice types—eMBB, URLLC, and mMTC—are instantiated as isolated logical networks atop this virtualised foundation. The MANO framework governs the entire slice lifecycle.

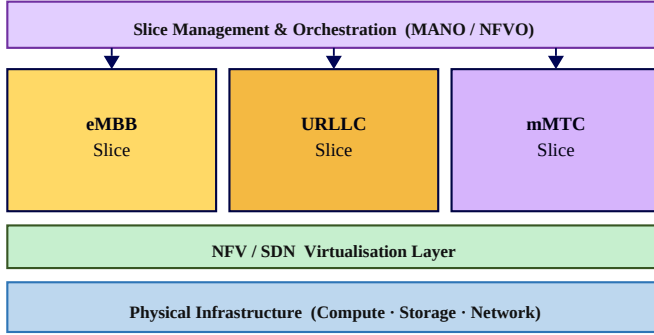


Fig. 1. Layered network slicing architecture in 5G NR.

Within the NG-RAN, slicing support is realised through the disaggregated gNB architecture, which separates the base station into a Central Unit (CU) and one or more Distributed Units (DUs). The CU hosts the higher-layer protocols (PDCP, RRC) and can be further split into a CU-CP (Control Plane) and a CU-UP (User Plane), enabling independent and elastic scaling of control and user-plane functions on a per-slice basis. Resource scheduling at the DU enforces per-slice radio resource budgets through the Radio Resource Management (RRM) entity, as specified in 3GPP TS 38.300 [2]. The Xn interface between gNBs supports inter-cell coordination for slice-aware mobility management.

IV. SLICE LIFECYCLE MANAGEMENT

The lifecycle of a network slice is defined by four distinct phases: preparation, commissioning, operation, and decommissioning. Effective management across all four phases is essential to meeting the stringent SLA requirements of each slice tenant while maintaining efficient use of shared physical resources.

During the *preparation* phase, the operator defines a Network Slice Template (NST) that encapsulates the slice's required capacity, topology, constituent VNFs, and QoS targets. This template serves as the blueprint for all subsequent instantiation steps. The *commissioning* phase involves the actual instantiation and configuration of all VNFs forming the slice's service function chain, the allocation of virtualised compute and radio resources, and end-to-end connectivity verification.

The *operational* phase is the longest-running and most complex. It encompasses continuous performance monitoring against SLA thresholds, automated elastic scaling of VNF instances in response to changing traffic load, and dynamic resource reallocation between slices via closed-loop control mechanisms. AI/ML-driven traffic prediction is increasingly being integrated at this stage to enable proactive, anticipatory resource management before SLA violations can occur [6]. Finally, the *decommissioning* phase gracefully terminates a slice instance, migrating active sessions where necessary and releasing all allocated resources back to the shared pool.

The ETSI NFV Management and Orchestration (MANO) architectural framework provides the standardised tooling for automating slice lifecycle management. The MANO stack consists of three components: the NFV Orchestrator (NFVO), responsible for cross-domain resource orchestration and slice composition; the VNF Manager (VNFM), responsible for the lifecycle of individual VNF instances; and the Virtualised Infrastructure Manager (VIM), responsible for managing the underlying virtualised compute, storage, and network resources.

V. USE CASES AND PERFORMANCE ANALYSIS

3GPP defines three canonical slice types, each targeting a distinct class of applications with fundamentally different QoS requirements. Understanding the performance profile of each slice type is essential to appreciating why a one-size-fits-all network design is inadequate and why isolation between slices is a strict requirement rather than a desirable feature.

A. Enhanced Mobile Broadband (eMBB)

eMBB slices are designed to deliver the highest possible data throughput to mobile users and fixed wireless access customers. The ITU-R IMT-2020 specification mandates peak downlink data rates exceeding 20 Gb/s and minimum guaranteed user-experienced rates of 100 Mb/s in dense urban deployments. These slices prioritise spectral efficiency, leveraging 5G NR capabilities such as massive MIMO antenna arrays with beamforming, millimetre-wave (mmWave) spectrum utilisation, and flexible carrier aggregation across multiple frequency bands. Because eMBB applications tolerate latencies of tens of milliseconds, UPF instances can be deployed centrally without adverse impact on user experience.

B. Ultra-Reliable Low-Latency Communications (URLLC)

URLLC slices target mission-critical applications including autonomous vehicle coordination, remote robotic surgery, real-time factory automation, and smart grid protection systems. These applications demand end-to-end latencies below 1 ms and packet error rates below 10^{-5} . Achieving these targets simultaneously requires a suite of 5G NR mechanisms: MEC-based UPF placement at the cell site or aggregation node to minimise transport delay; mini-slot scheduling with transmission time intervals as short as 2 OFDM symbols; grant-free uplink transmission to eliminate scheduling request round trips; and HARQ retransmission with tight timing constraints [5].

C. Massive Machine-Type Communications (mMTC)

mMTC slices are engineered to support the massive scale of IoT deployments, targeting connection densities of up to 10^6 devices per km². These devices are characterised by sporadic, low-volume uplink transmissions, extremely long battery lifetimes measured in years, and a tolerance for latencies of seconds to minutes. The 5G NR feature set for mMTC includes Extended Discontinuous Reception (eDRX) and Power Saving Mode (PSM) to maximise device sleep time, NB-IoT and mMTC waveform support for deep building penetration, and lightweight non-access stratum (NAS) procedures to minimise signalling overhead for massive device populations.

D. Comparative QoS Analysis

Figure 2 presents a normalised comparison of three key QoS dimensions across the three standard slice types: throughput, inverse latency (where a higher score indicates lower absolute latency), and therefore better performance for latency-sensitive applications), and reliability. The stark contrast between the profiles of eMBB, URLLC, and mMTC underscores why slice isolation is a fundamental architectural requirement. A resource configuration optimised for

eMBB throughput would impose catastrophic latency on URLLC industrial control traffic. Conversely, mMTC requires neither the bandwidth of eMBB nor the strict latency of URLLC, but demands connection density and energy efficiency that neither of the other slice types is designed to provide.

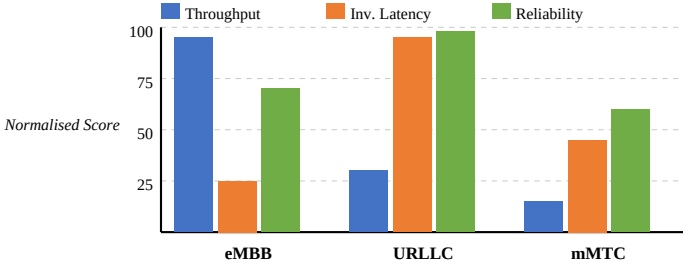


Fig. 2. Normalised QoS profile comparison across eMBB, URLLC, and mMTC slices.

VI. OPEN RESEARCH CHALLENGES

Despite the significant standardisation progress achieved in 3GPP Releases 15 through 17, a number of important research and engineering challenges must still be addressed before network slicing can be fully realised across all deployment scenarios.

Radio-Layer Isolation: Inter-slice isolation guarantees are relatively straightforward to enforce in the core network through VNF isolation and virtual networking. However, at the radio interface, the shared wireless channel introduces physical-layer coupling between slices that share the same spectrum. Dynamic admission control policies must balance the acceptance of new slice requests against the protection of the SLA commitments of already-admitted tenants, without resorting to excessive over-provisioning of radio resources [7].

Security and Isolation: Slice-specific security policies must rigorously prevent cross-slice attacks and data exfiltration while simultaneously permitting legitimate inter-slice communication for applications that consume services from multiple slice types. Furthermore, end-to-end slicing across multi-operator and multi-administrative-domain scenarios requires standardised northbound APIs, federated trust models, and slice peering agreements that are still under active development in 3GPP Release 18 and ETSI working groups.

AI/ML-Driven Management: The integration of machine learning for autonomous, closed-loop slice management—encompassing traffic demand forecasting, anomaly detection, and multi-objective resource optimisation—introduces a new set of challenges. These include the generalisability of trained models to previously unseen traffic patterns, the privacy implications of using subscriber data for model training, and the inference latency of deployed models, all of which must be carefully addressed before these techniques can be safely deployed in production 5G networks.

VII. CONCLUSION

This paper has presented a comprehensive overview of network slicing in 5G New Radio, from its foundational enabling technologies to its standardised architecture, lifecycle management framework, and canonical use cases. We demonstrated that SDN, NFV, and MEC collectively provide the programmability, virtualisation, and edge proximity needed to instantiate isolated, customised end-to-end slices atop a shared physical infrastructure.

The analysis of eMBB, URLLC, and mMTC slice types, supported by a comparative QoS profile evaluation, confirmed that the radically different performance requirements of these service classes cannot be simultaneously satisfied by any single network configuration, thereby validating network slicing as an architectural necessity rather than a convenience feature. While network slicing is now being commercially deployed in 5G networks worldwide, open challenges in radio-layer isolation, multi-domain interoperability, and AI-driven autonomous management continue to present rich avenues for future research. Addressing these challenges is essential to realising the full promise of 5G as a truly universal, programmable connectivity platform for the decade ahead.

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